ARNAV SHARMA

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EDUCATION

Indian Institute of Technology (IIT) Jodhpur Aug 2024 – May 2028 (Expected)

B.Tech in Mechanical Engineering

Relevant Coursework: Machine Learning, Probability & Statistics, Linear Algebra, Data Structures & Algorithms

RESEARCH INTERESTS

Adversarial Robustness & Security of Tool-Using LLM Agents, Runtime Information-Flow / Taint-Tracking Guardrails, Programmatic Verification of Autonomous Data-Science Pipelines, and Reliable Tabular Inference.

RESEARCH & SYSTEMS PROJECTS

Agent Red-Team (github.com/arnavsharma09/agent-redteam) *Python, Ollama (Qwen2.5-7B), UCB1 Bandit, Wilson CIs, pytest*

- Built an adaptive red-team harness for tool-calling LLM agents: a UCB1 bandit searches 144 prompt-injection recipes against a local Qwen2.5-7B agent; undefended, injections succeeded in 17.4%–37.0% of episodes (37.0% destructive, 17.4% exfiltration over 350 episodes).
- Implemented an argument-provenance taint guardrail: exfiltration succeeded in 0/250 episodes vs 61/350 undefended, with 0/36 benign-task blocks (Wilson 95% upper bound 9.6%) and 35/36 task completion.
- Closed code-execution and base64 self-decoding evasion gaps (0/100 executions after fix); 179 unit tests passing.

AutoInsight (github.com/arnavsharma09/AutoInsight) *LangGraph, Polars, ChromaDB, FastAPI, Streamlit, Anthropic Claude API, Docker*

- Architected a 7-node multi-agent pipeline (Profiler, Router, Planner, Coder, Sandbox, Critic, Reporter) enforcing Shapiro-Wilk normality profiling and automated non-parametric fallback (e.g., ANOVA → Kruskal-Wallis + Dunn post-hoc), with a human-in-the-loop approval checkpoint.
- Sandboxed all generated code execution in an isolated Docker container with AST-level security scanning (blocking os, subprocess, eval), and validated report faithfulness against computed metrics using DeepEval.

RetainIQ (github.com/arnavsharma09/RetainIQ) *PostgreSQL, SQL, Python (scipy, statsmodels), Power BI, Docker*

- Built an end-to-end analytics pipeline on 99,441 orders across 9 relational tables (Brazilian e-commerce dataset), modeling a star schema in PostgreSQL and applying quintile-scored RFM segmentation across 94,983 customers into 7 value tiers.
- Ran a controlled hypothesis test isolating delivery time's effect on satisfaction independent of Haversine distance: Shapiro-Wilk confirmed non-normality, so Mann-Whitney U showed fast-delivery orders scored significantly higher (4.31 vs 3.75, $p < 0.001$).
- Identified a 3.04% repeat-purchase rate and 62.5% revenue concentration across 3 states; shipped a 6-visual interactive Power BI dashboard.

TECHNICAL SKILLS

- **Languages & Core Stack:** Python, SQL (PostgreSQL), C, C++, Bash, Linux, Git, Docker
- **LLM Security & Agentic Systems:** LangGraph, Ollama (qwen2.5:7b), UCB1 Bandit Search, Runtime Taint-Tracking, AST Static Analysis, DeepEval, ChromaDB
- **Statistical Inference & Tabular ML:** Polars, SciPy, statsmodels, scikit-learn, XGBoost, PyTorch, Non-Parametric Hypothesis Testing, Wilson CIs
- **Data Systems & Reproducibility:** pytest (179 test suite), FastAPI, SQLAlchemy, Dimensional Star-Schema Modeling

LEADERSHIP & POSITIONS OF RESPONSIBILITY

Assistant Head, Marketing - Prometeo 26 | IIT Jodhpur 2025–2026

- Led cross-functional outreach and digital sponsorship channels for IIT Jodhpur's flagship technical festival.

Student Guide - Student Wellbeing Committee | IIT Jodhpur 2025–2026

- Peer-support mentor guiding incoming first-year engineering students through academic and campus transitions.